Cut Sheet

Board Design Summary

Style: checkerboard

Dimensions: 300.0 mm × 450.0 mm × 40.0 mm

Species: Walnut, Maple

Juice Groove: Perimeter loop, 12.5 mm wide × 5 mm deep, 19 mm from the edge, corner radius 25.5 mm, top face.

Rubber Feet: 4 rubber feet, 1" (25.4 mm) diameter × 3/8" (9.5 mm) high, foot centres 32 mm in from the edges (underside).

[Step Diagram]

Cut List

Part	Qty	Width	Thickness	Length	Species
Walnut strip	108	34.2 mm	46.5 mm	78.2 mm	Walnut
Maple strip	108	34.2 mm	46.5 mm	78.2 mm	Maple

Board Layout (Top View)

[Step Diagram]

Step 1: Mill rough stock

Joint one face and one edge on the jointer, then plane to final thickness plus 1/8" (3 mm). Rest overnight, then plane to final thickness. Take ~1/64" (0.4 mm) passes for the last two passes.

Settings:

Final pass depth: ~1/64" (0.4 mm)

WARNING: Use push blocks on the jointer. Never joint stock shorter than 12" (305 mm) or thinner than 1/2" (12.5 mm).

WARNING: Keep hands clear of the infeed and outfeed. Feed only boards at least 12" (305 mm) long.

[Step Diagram]

Step 2: Rip strips

Rip strips with the jointed edge against the fence. Measure from the fence face to the near side of a tooth set toward the fence.

Settings:

Fence to blade: strip width from cut list

WARNING: Use the riving knife and blade guard on every through cut.

WARNING: Use a push stick for any rip narrower than 6" (152 mm). Never reach over or behind the blade.

[Step Diagram]

Step 3: Glue up

Glue the first edge-grain panel. Alternate clamps above and below the panel and check for flat with a straightedge.

Settings:

Clamp spacing: about 8" (200 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

[Step Diagram]

Step 4: Flatten first panel

Plane or sand the panel flat. It is still long grain, so the planer is allowed here.

WARNING: Keep hands clear of the infeed and outfeed. Feed only boards at least 12" (305 mm) long.

[Step Diagram]

Step 5: Crosscut slices

Crosscut the panel into slices of the final thickness on a crosscut sled with a stop block.

Settings:

Slice length: final board thickness from cut list

DANGER: Never use the rip fence as a length stop for crosscuts. Use a crosscut sled with a stop block set before the blade, so the offcut is never trapped between blade and fence.

WARNING: Do not cut pieces shorter than 6" (152 mm) free-hand on the table saw. Use a sled with a hold-down.

[Step Diagram]

Step 6: Rotate slices

Turn each slice end grain up.

Step 7: Flip alternate slices

Flip every other slice end for end so the colours alternate into squares.

Step 8: Glue up

Glue slices into the final panel, aligning the squares at every joint. Alternate clamps above and below the panel and check for flat with a straightedge.

Settings:

Clamp spacing: about 8" (200 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

[Step Diagram]

Step 9: Flatten and sand

Flatten with a router sled or drum sander, never the thickness planer. Sand 80, 120, 180 grit. Break edges (180 grit).

Settings:

Grits: 80 / 120 / 180

DANGER: Never run an end-grain glue-up through the thickness planer. It can blow apart. Flatten with a router sled or drum sander.

[Step Diagram]

Step 10: Edge profile

Rout the edge profile, default: 1/8" (3 mm) roundover. Optional finger grips: core box / round-nose bit 3/8–1/2" (10–12 mm).

Settings:

Bit: 1/8" (3 mm) roundover

WARNING: Climb cuts can grab. Feed the router against the bit rotation and take shallow passes.

Step 11: Finish

Raise the grain with a damp cloth, sand 180 again, then flood with food-safe mineral oil and top with board butter.

NOTE: Use closed-grain, non-toxic hardwoods only. Avoid spalted, softwood or oily/sensitising exotics on cutting surfaces.